CHAPTER 2

THEORY OF CONSUMER BEHAVIOUR

CHAPTER ESSENCE AT A GLANCE:
Analyze rational consumer choice through Cardinal Utility (Law of DMU) and Ordinal Utility (Indifference Curves and Budget Line), deriving demand curves and price elasticity.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Utility and Law of Diminishing Marginal Utility:** Utility measures want-satisfying power; the Law of DMU states that as more units of a good are consumed, the marginal utility derived from each successive unit decreases.

• **Indifference Curve Convexity and MRS:** An Indifference Curve shows bundles yielding equal satisfaction; its convex shape reflects diminishing Marginal Rate of Substitution (MRS) as the consumer sacrifices less of good 2 for each unit of good 1.

• **Budget Constraint and Budget Line:** The budget line represents all consumption bundles costing exactly the consumer's income (p1 x1 + p2 x2 = M); its slope equals the negative price ratio (-p1 divided by p2).

• **Consumer Equilibrium Tangency Condition:** Equilibrium occurs where the highest attainable indifference curve is tangent to the budget line, satisfying MRS = p1 divided by p2, with the indifference curve being convex to the origin.

ESSENTIAL ECONOMIC MODELS & GOVERNING FORMULAS

$$\text{Consumer Equilibrium (Ordinal)} \iff \text{MRS}_{xy} = \frac{p_x}{p_y} = \frac{\text{MU}_x}{\text{MU}_y}$$

Tangency of Indifference Curve and Budget Line

$$\text{Budget Line Equation} \iff p_1x_1 + p_2x_2 = M \rightarrow \text{Slope} = -\frac{p_1}{p_2}$$

Price Ratio Governing Expenditure Boundary

$$\text{Marginal Rate of Substitution (MRS)} = -\frac{\Delta x_2}{\Delta x_1} \rightarrow \text{Diminishing Convex Slope}$$

Willingness to Substitute at Constant Satisfaction

$$e_D = -\frac{\%\Delta Q}{\%\Delta P} = -\frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

Price Elasticity of Demand Percentage Formula

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Never confuse a change in quantity demanded (movement along demand curve due to own price change) with a change in demand (shift of demand curve due to income or tastes).

• **Alert:** At consumer equilibrium, the condition MRS = p1 divided by p2 is necessary but not sufficient; the indifference curve must also be convex to the origin at that point.

• **Alert:** Remember that Price Elasticity of Demand is conventionally written as a positive number by prefixing a minus sign to neutralize the inverse price-quantity relationship.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. State the Law of Diminishing Marginal Utility with its two essential assumptions. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Statement:** Other things remaining constant, as a consumer consumes more and more units of a commodity, the marginal utility derived from each additional unit continuously decreases. **[1 Mark]**

• **Assumptions:** (1) Standard and reasonable unit size of commodity consumed; (2) Continuous consumption without time lag. **[1 Mark]**

2. Explain the relationship between Total Utility (TU) and Marginal Utility (MU) with the help of a schedule. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Schedule:** Units (1 to 5): MU = 10, 8, 4, 0, -2. TU = 10, 18, 22, 22, 20. **[1 Mark]**

• **Core Relationships:**
• When μ is positive, TU increases at a diminishing rate.
• When μ is zero, TU reaches its maximum (Point of Satiety). **[1 Mark]**

• • When MU becomes negative, TU begins to diminish. **[1 Mark]**

3. Explain any four essential properties of Indifference Curves. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **1. Downward Sloping to the Right:** To remain on the same satisfaction level, an increase in consumption of good 1 must be accompanied by a decrease in consumption of good 2. **[1 Mark]**

• **2. Strictly Convex to the Origin:** Due to the diminishing Marginal Rate of Substitution (MRS), the consumer is willing to give up fewer units of good 2 for each additional unit of good 1. **[1 Mark]**

• **3. Higher IC Represents Higher Satisfaction:** Based on monotonic preferences, a higher indifference curve contains more of at least one good and no less of the other. **[1 Mark]**

• **4. Two ICs Never Intersect:** If two curves intersected, they would imply contradictory levels of utility for the same common commodity bundle. **[1 Mark]**

4. Explain consumer equilibrium under Indifference Curve analysis using conditions of equilibrium. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Condition 1 (Tangency):** MRS equal to the price ratio: MRS = p1 divided by p2. The slope of the indifference curve must equal the slope of the budget line. **[1.5 Marks]**

• **Condition 2 (Convexity):** The indifference curve must be strictly convex to the origin at the point of tangency (diminishing MRS). **[1 Mark]**

• **Disequilibrium Adjustment:** If MRS > p1 divided by p2, the consumer values good 1 more than market price, buying more good 1, reducing MRS until equality is restored. If MRS < p1 divided by p2, the consumer buys more good 2 until equilibrium is reached. **[1.5 Marks]**

5. A consumer buys 50 units of a good at price Rs 10 per unit. When the price falls to Rs 8 per unit, the quantity demanded rises to 70 units. Calculate price elasticity of demand. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Given Data:** Initial Price P = 10, New Price P1 = 8. Delta P = 8 - 10 = -2. Initial Quantity Q = 50, New Quantity Q1 = 70. Delta Q = 70 - 50 = +20. **[1 Mark]**

• **Formula:** eD = - (Delta Q divided by Delta P) * (P divided by Q). **[1 Mark]**

• **Calculation:** eD = - (+20 divided by -2) * (10 divided by 50) = 10 * 0.2 = 2.0. The demand is highly price elastic (eD > 1). **[1 Mark]**

6. What is a budget line? Write its equation and state the formula for its slope. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** A budget line is a graphical representation of all possible combinations of two goods that a consumer can purchase with their entire money income given market prices. **[1 Mark]**

• **Equation and Slope:** Equation is p1 x1 + p2 x2 = M. Slope = - (p1 divided by p2), representing the market rate of exchange between the two commodities. **[1 Mark]**